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3 Complex Analysis 2023

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Test 2

11 October 2023



Instructions:

- This question paper consists of **6 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 40 points = 100% (42 points available).

Problem	1	2	3	4	5	Σ	Mark
Maximum Points	8	10	8	8	8	40	100%
Result							

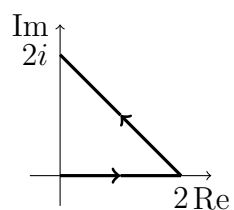
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Problem 1

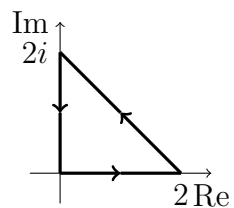
(8 points)

Calculate the following integrals.

- (a) $\int_{\gamma} z^2 dz$, where γ is the indicated path:



- (b) $\int_{\gamma} z^2 dz$, where γ is the indicated path:



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Problem 2

(12 points)

Use Cauchy's Integral Formula and Cauchy's Integral Formula for derivatives to calculate the following integrals.

(a) $\int_{\gamma} \frac{\cos \pi z}{z^2 - 1} dz$, where $\gamma(t) = 3e^{it}$, $t \in [0, 2\pi]$.

(b) $\int_{\gamma} \frac{e^z}{(z - 2)^2} dz$, where $\gamma(t) = 2 + e^{it}$, $t \in [0, 2\pi]$.

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Problem 3

(8 points)

- (a) Formulate the Coincidence Principle for functions f and g that are holomorphic in a region $G \subseteq \mathbb{C}$.
- (b) Does there exists a function f that is holomorphic in the open disc $D(0; 3) = \{z \in \mathbb{C} : |z| < 3\}$ such that

$$f\left(\frac{n-1}{n}\right) = f\left(\frac{n+1}{n}\right) = \frac{1}{n} \quad \text{for all } n \in \mathbb{N}?$$

Justify your answer.

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Problem 4

(8 points)

Show that there exists no holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ with

$$|f'(0)| > \max_{|z|=1} |f(z)|.$$

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Problem 5

(8 points)

Find all singularities of the function $f(z) = \frac{(z+1)(z-i)}{(z^2+1)(z+i)}$ and determine their type (a removable singularity, a pole, or an essential singularity). For a removable singularity, give a holomorphic extension of f . For a pole, give its order.